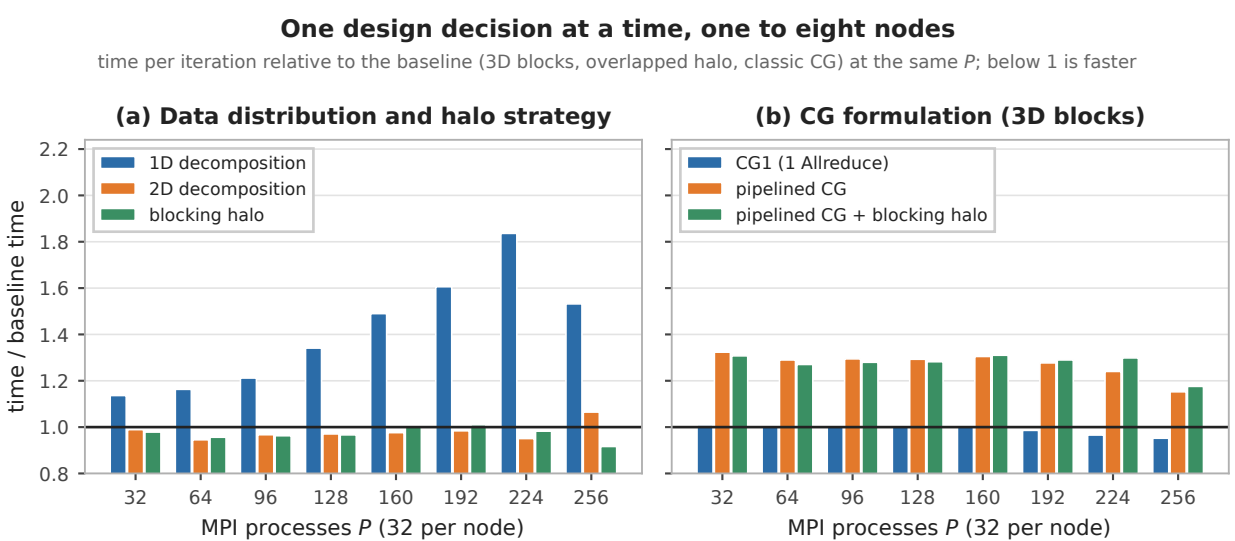
One design decision at a time, one to eight nodes
time per iteration relative to the baseline (3D blocks, overlapped halo, classic CG) at the same P; below 1 is faster
(a) Data distribution and halo strategy
(b) CG formulation (3D blocks)
time / baseline time
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
1D decomposition
2D decomposition
blocking halo
CG1 (1 Allreduce)
pipelined CG
pipelined CG + blocking halo
32
64
96
128
160
192
224
256
MPI processes P (32 per node)
MPI processes P (32 per node)